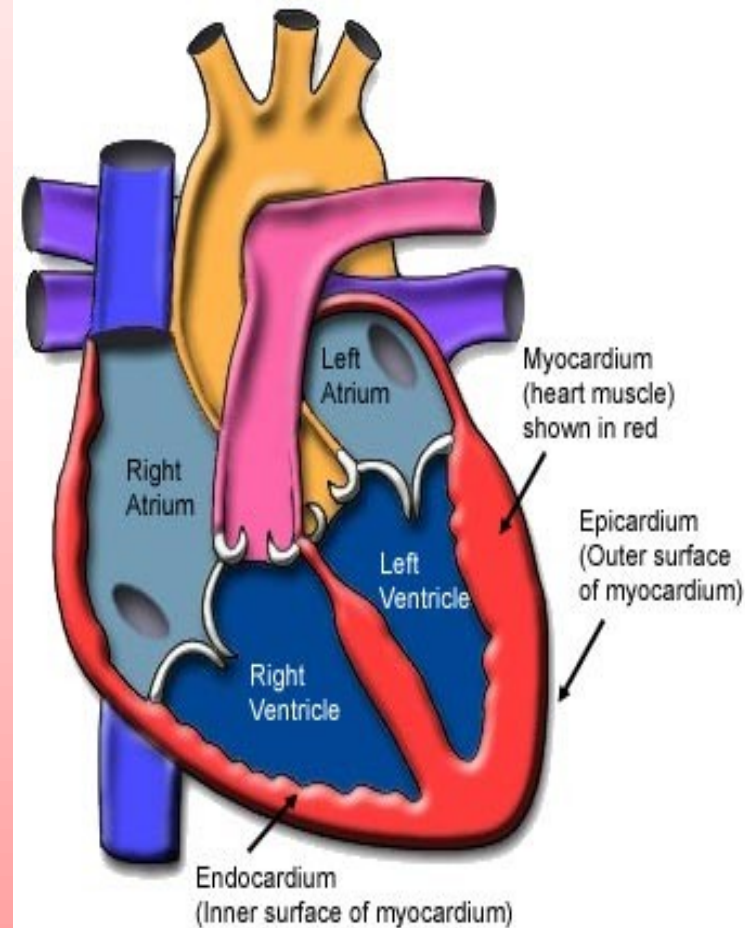


Module title:

Cardiovascular & lymphatic system

- Microbes affecting the heart & pericardium



Microbes affecting the heart & pericardium

- **Pericardium**

- ⇒ Fibrous membrane surrounding heart

- **Heart layer**

- ⇒ Endocardium

- ✓ The inner layer (valve & other inner structures)

- ⇒ Myocardium

- ✓ Thickest & contractile layer

- ⇒ Epicardium

- ✓ Outer coat lubricating the heart

Cont...

- Different microbes can affect the heart and pericardium
 - ⇒ Bacteria, Viruses, Parasites, Fungi
- Common heart & pericardium disease → by microbial agent
 - ⇒ Infective endocarditis → inflammation of endocardium by infectious agents
 - ✓ Acute infective endocarditis
 - ✓ Sub acute infective endocarditis
 - ⇒ Myocarditis
 - ⇒ Pericarditis

Microbial Diseases of the CVS

□ Diseases of the CVS

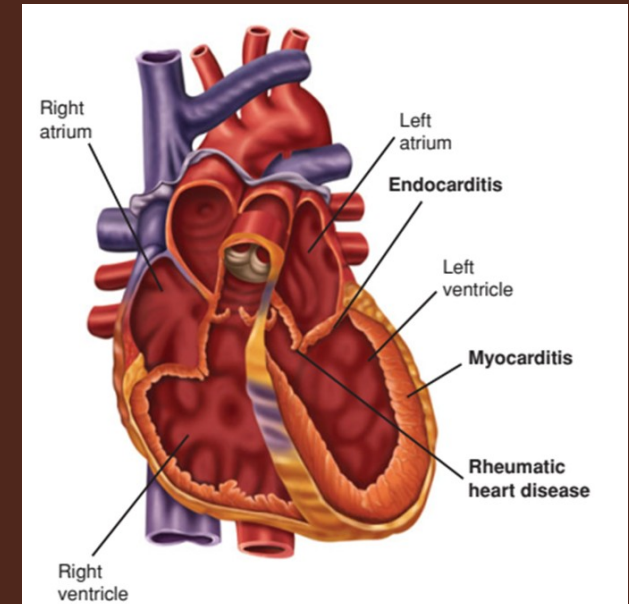
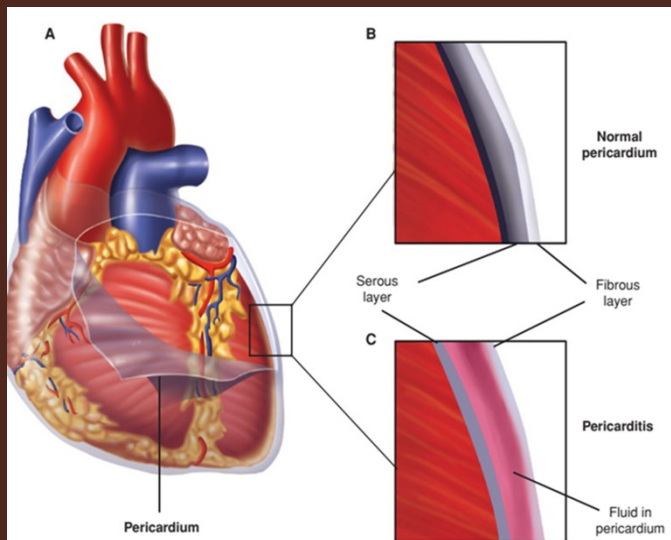
Infective endocarditis (IE)

Prosthetic valve endocarditis (PVE)

Rheumatic fever

Pericarditis

Myocarditis



□ Microbiology of Etiologic Agents

Native and Prosthetic valve endocarditis,

S. aureus

Coagulase-negative staphylococci

Streptococci (+ *viridans*)

Enterococci

Pseudomonas + Gram-negative rods (HACEK)

Candida species

Pericarditis

Streptococcus pneumoniae

S. aureus

M. tuberculosis

Coxsackie virus B

Myocarditis

Corynebacterium diphtheriae

Borrelia burgdorferi

Trypanosoma cruzi

Coxsackie group B (less often A), enteroviruses,
mumps, influenza and rubella

Endocarditis

An inflammation of the inner surface, or endothelium of the heart.

The surfaces most commonly affected are the surfaces of the heart valves (native or prosthetic)

Classified as infective or noninfective

Noninfective endocarditis is very rare and can follow autoimmune reactions or can occur in patients with mucinous adenocarcinoma

prototypic lesion of infective endocarditis is the vegetation

- A mass of platelets, fibrin, microcolonies of microorganisms, and scant inflammatory cells.

Infective Endocarditis ...

■ IE has been classified by three major factors:

1. **Severity and rapidity of infection**
 - Acute, developing in days, (often 3–6)
 - Subacute, in weeks, (often 3–6)
 - Chronic, over months (often 3 or more)
2. **Valve type**
 - Native valve of the patient
 - Prosthetic valve
3. **Source of bacteraemia, if known**
 - dental, medical or surgical procedures
 - drugs: injecting drug use (IDU or IVDU)
 - devices, e.g., intravascular catheters
 - diseases elsewhere, e.g. abscesses.

Infective Endocarditis ...

Embolization, especially in Gram-positive and fungal infection, produces distal infarction in

the periphery [splinter hemorrhages under finger or toe nails, or skin or conjunctival petechiae – Roth spots]

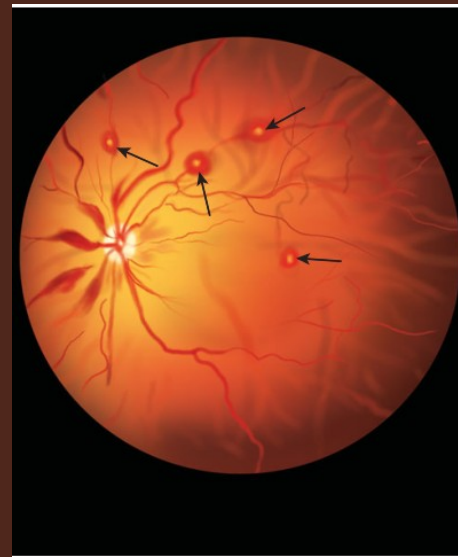
vascular organs, including the spleen, kidney or brain.

embolization also causes mycotic aneurysms in arteries or distal abscesses including micro abscesses ('Janeway lesions').

Immune complex deposition causes glomerulonephritis commonly, and Osler's nodes rarely; the latter are small painful red lumps in fingertips, palms or soles.



A photograph of splinter haemorrhages under finger nails



Roth spots (arrows) present on the retina of a patient with endocarditis

○Subacute bacterial endocarditis

commonly due to *viridans streptococci*, enterococci, staphylococci or unusual Gram-negative rods.
causes weeks of fever, emboli, immunologic phenomena and cardiac murmur(s).

○Chronic endocarditis (rare)

Due to fungi, Q fever or coagulase-negative staphylococci.
causes months of fever, splenomegaly and murmur(s).

Infective Endocarditis ...

○Prosthetic endocarditis

occurs either

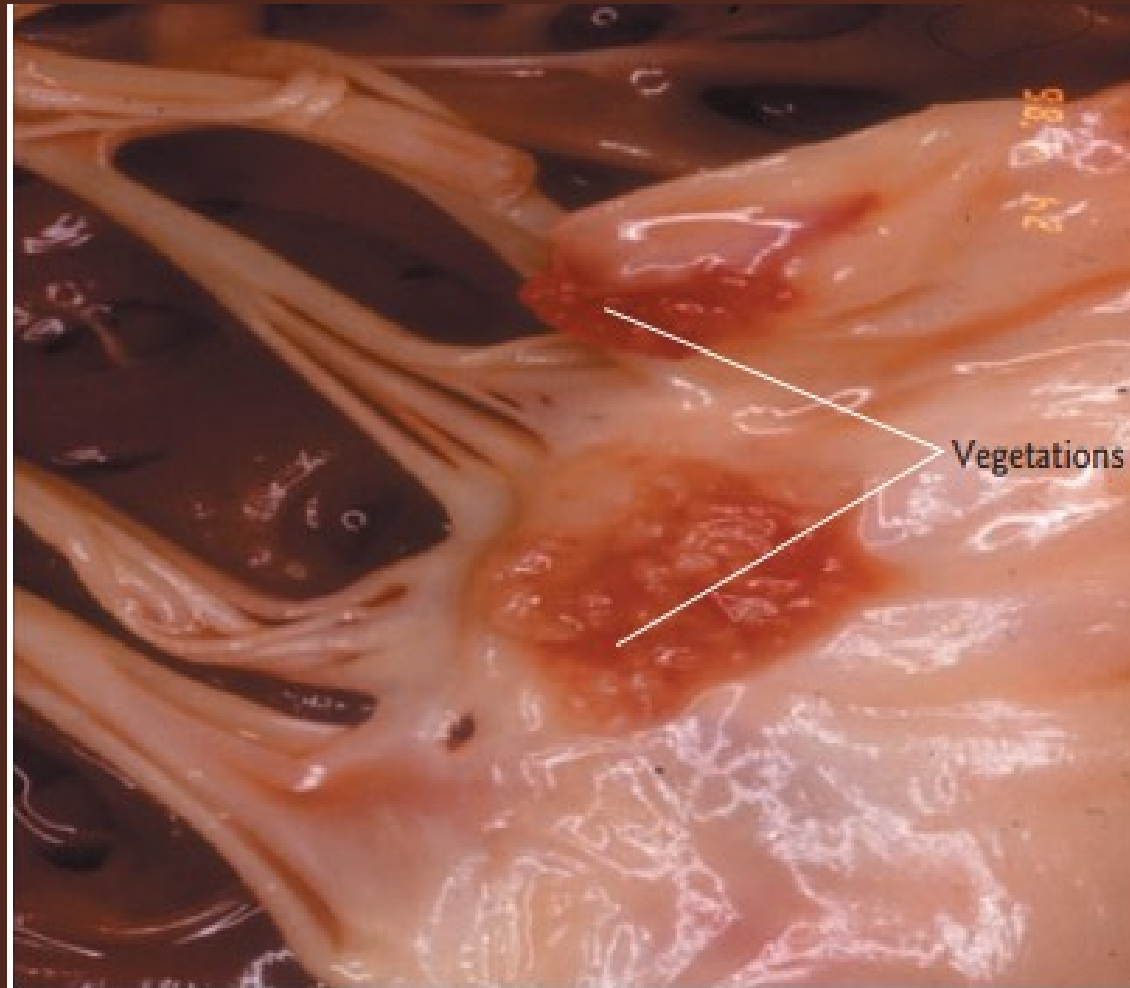
early, after operation usually due to *Staphylococci* or enteric GNRs

late, 2 months or more after operation, usually due to oral *Streptococci*, *Enterococci* or *coagulase-negative staphylococci*

Fever, bacteraemia and a new murmur occur earlier than emboli.

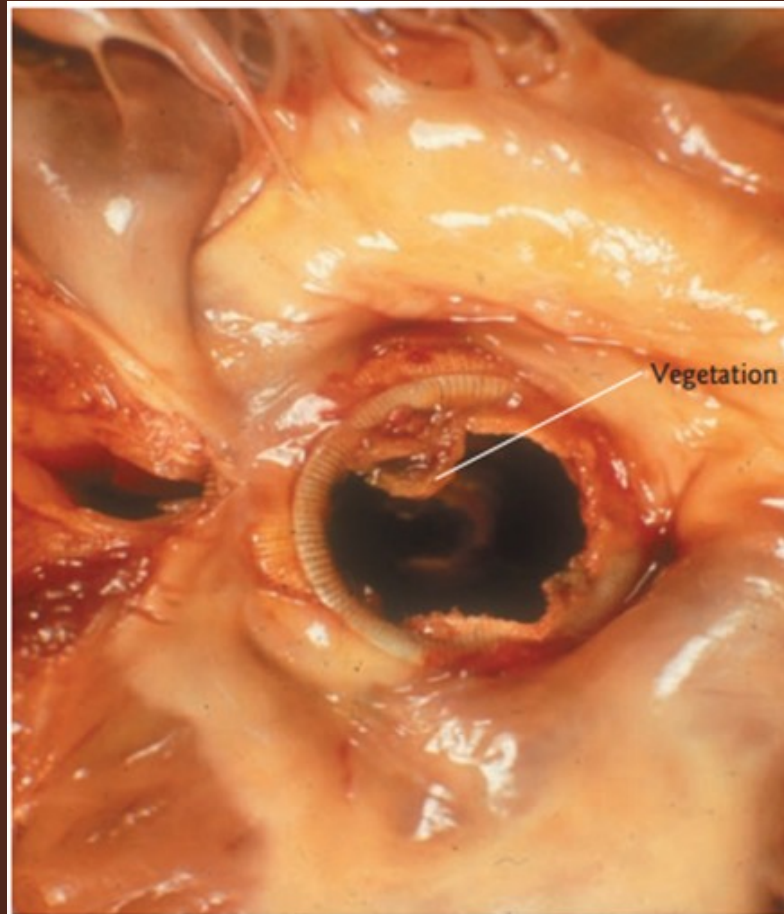
Perivalvular leaks are serious; an occluded valve is rapidly fatal.

Infective Endocarditis ...



Native aortic valve with bacterial vegetations

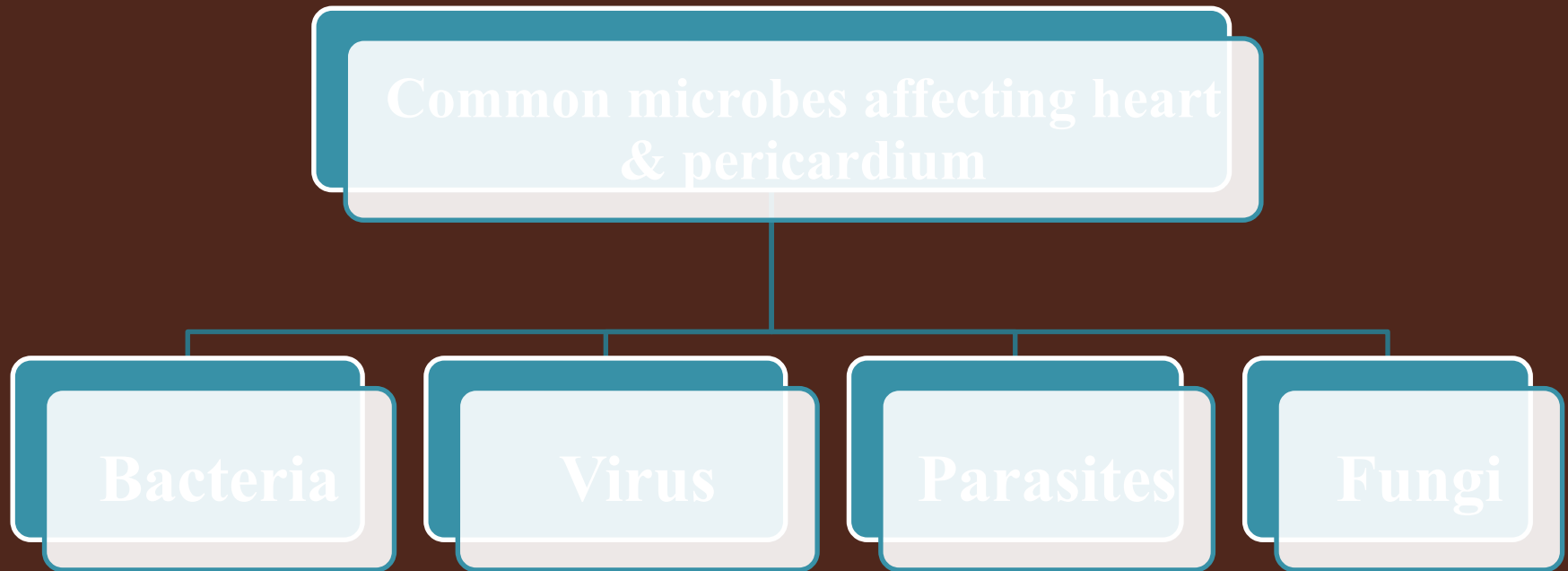
Infective Endocarditis ...



Prosthetic valve endocarditis

Cont...

Chapter overview



1. Bacteria

1.1. *Staphylococcus aureus*

- Gram positive cocci
- Have d/t structural components
 - ⇒ Capsule, protein-A (bind Ab-prevent com activation)
 - ⇒ Surface proteins
- Can survive for long period
 - ⇒ Thick peptidoglycan layer
- Normal flora
 - ⇒ Skin, nasopharynx ($\geq 15\%$)

Cont...

❖ Pathogenesis

- **Transmission** → to a susceptible person
 - ⇒ Direct contact
 - ⇒ Implantation → E.g. medical equipments
- **Virulent factors**
 - ⇒ **Bacterial structure** → surface structure & capsule
 - ✓ Mediate adherence → to host tissues
 - ▶ Including **valve endothelial** cells
 - ✓ Antiphagocytic
 - ✓ Prevent complement activation
 - ▶ Protein A → bind Fc portin of Ab

Cont...

- ⇒ **Extra cellular product** → proteins
 - ✓ **Toxins & enzymes**
 - ▶ Exfoliative, enterotoxins & TSST toxins
 - ▶ Coagulase, Catalase, Hyaluronidase, Fibrinolysin

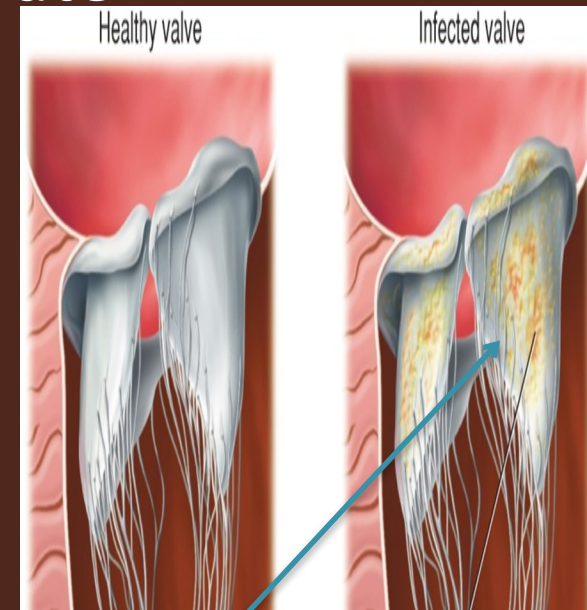
Cont...

❖ Diseases

- Cause a wide Variety of disease
 - ⇒ Go beyond superficial infections
 - ✓ Traumatic injury, immunosuppression and other factors are some of the risk factors
- Bacteremia
 - ⇒ *S. aureus* → the common cause
 - ✓ From → **known/unknown** focus of infection

Cont...

- Prolonged bacteremia
 - ⇒ Cause endocarditis (*usually acute*)
 - ✓ Commonly on native valve
 - ⇒ Serious disease → ~50% mortality rate
- The disease is due to
 - ⇒ Bacterial proliferation
 - ✓ Attach & establish growth
 - ⇒ Deposition occurs → fibrin & platelet
 - ✓ Interfere phagocytosis & complement
 - ⇒ Few inflammatory cells



■ Prototypic lesion at site of infection called → **vegetation**

Cont...

- Can also lead to abscess formation & tissue destruction
 - ⇒ Valve destruction → may cause heart failure
 - ⇒ Can spread to myocardium

❖ Laboratory diagnosis

- ⇒ Culture
- ⇒ Molecular technique → PCR

Cont...

1.2. *Staphylococcus epidermidis*

- Gram positive cocci
 - ⇒ Arranged in clusters
- Normally flora
 - ⇒ Skin & mucosa (nose),
- Can survive for long period
 - ⇒ Thick peptidoglycan layer

Cont...

❖ Pathogenesis

■ Transmission

⇒ Mostly endogenous source

✓ Implantation → d/t medical device

✓ Manipulations → causing bacteremia

■ Virulent factor → Have surface structures

⇒ Slime layer → Mediate adherence

✓ Viscous exopolysaccharide intercellular adhesion (PIA)

➤ A multi-layered biofilm

⇒ Not as much as *S. aureus*

Cont...

- Commonly affects valves
 - ⇒ Previously damaged or
 - ⇒ Artificial heart valve
 - ✓ May be introduced → during replacement
 - ✓ Patients with foreign body → ↑ risk
 - ⇒ Slowly develop sign & symptoms → >1year

Cont...

❖ Diseases

- Endocarditis
- Artificial valve → if abscess formed
 - ⇒ Suture separation may occur
 - ✓ Lead to heart failure

❖ Laboratory diagnosis

- Culture
- Molecular technique → PCR

Cont...

- individuals with damaged/prosthetic valves have ↑risk endocarditis
- **Colonization** by circulating bacteria
 - ⇒ Low in normal valve endothelium
- **Disruption** of endothelium
 - ⇒ Expose extracellular matrix protein
 - ⇒ **Deposition** of fibrin & platelet
 - ✓ As normal healing process
- This deposition facilitate → **adherence & infection of bacteria** using their d/t strictures to attach to the area
- Endothelial damage can be caused by different agents like
 - ⇒ Mechanical lesion → E.g. turbulent b'd flow
 - ⇒ Catheters, rheumatic fever

Cont...

1.3. Streptococcus pyogenes

- Gram +ve, group A streptococcus
- Capsulated
- Part of normal flora
 - ⇒ Mainly in skin & oropharynx
- Survive on dry surfaces → for long periods
 - ⇒ Thick peptidoglycan layer

Cont...

❖ Pathogenesis

■ Transmission

⇒ Failure of non-specific immune system → in oropharynx

⇒ Transfer to susceptible person → gaining new strain

- ✓ By respiratory droplets

- ✓ Direct contact

 - ▶ Through skin breaks

- ✓ Fomite

Cont...

- **Virulent factors**

- ⇒ Important for pathogenesis

- ⇒ Bacterial & extra cellular factors

- **Bacterial structures**

- ⇒ Have different structures used for adherence

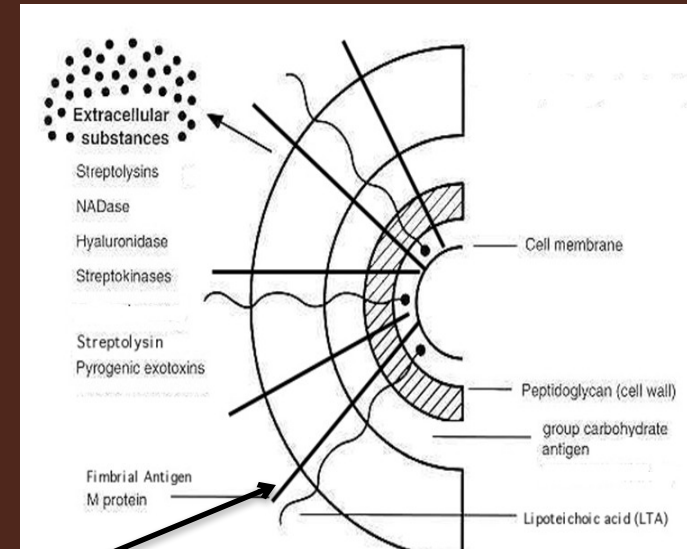
- ⇒ M-Protein is hair like projection

- ✓ Affect complement activation

- ▶ Bind Fc portion of antibody

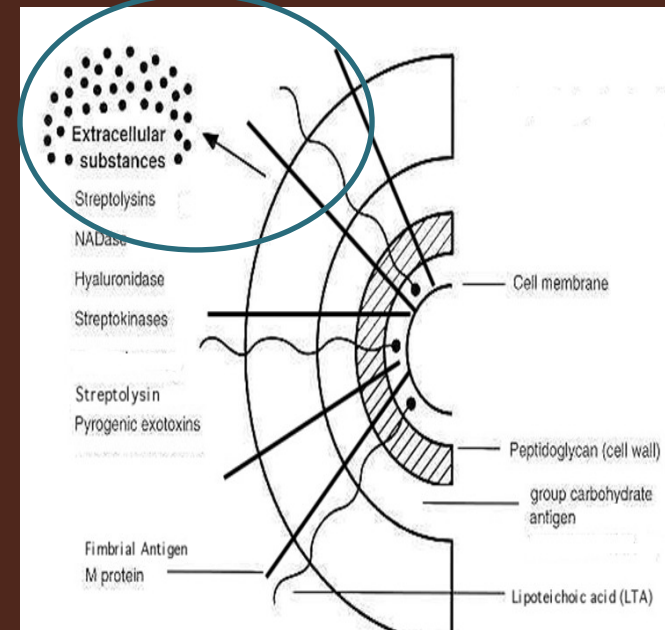
- ✓ >100 antigenically d/t types

- ▶ Cross reactive



Cont...

- **Extra cellular product** → Enzymes, hemolysin & toxin
- Some of these include:
 - ⇒ DNase (A-D)
 - ✓ Spread in tissue → by degrading extra cellular DNA
 - ⇒ Streptolysin O
 - ✓ Antigenic → Ab will be produced
 - ✓ Lyses b'd cells
- The bacteria initiate local infection
 - ⇒ Skin & respiratory tract
 - ⇒ Can spread to other organs



Cont...

❖ Diseases

A. Acute rheumatic fever

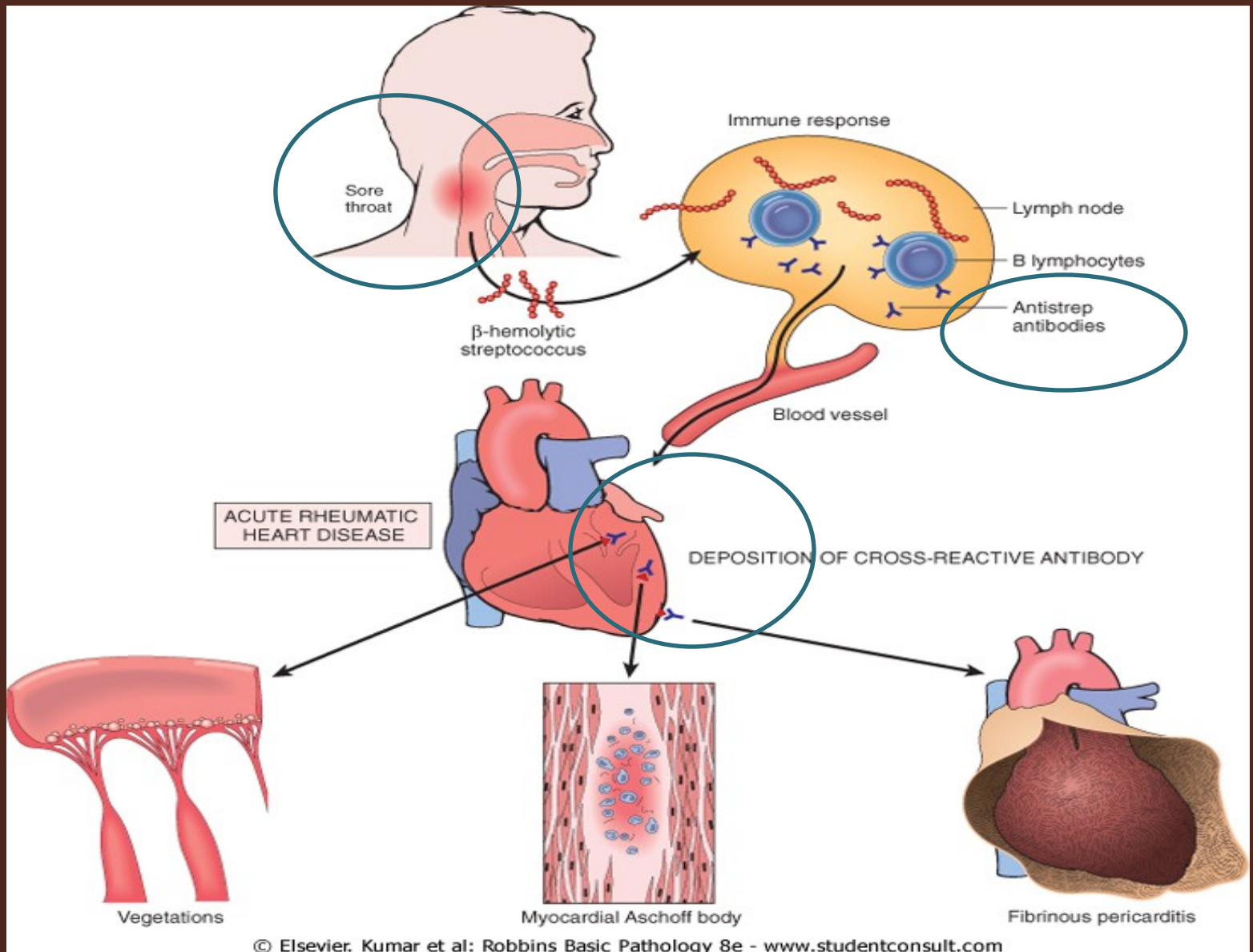
- ⇒ Post streptococcal infection → usually after 3 week of infection
- ⇒ Not directly associated with bacterial dissemination
- Occur because of antigenic mimicry
 - ⇒ M protein with Myocardial cell (myosin)
 - ⇒ Group-A carbohydrate with Valve endothelium

Cont...

- The disease
 - ⇒ Is associated with → streptococcal pharyngitis
 - ⇒ Not related to skin infections
 - ✓ No clear explanation
 - ⇒ Not occur in all pharyngitis
 - ✓ Have genetic factor
- Most common in school children & developing countries

Cont...

- Acute rheumatic fever
 - ⇒ Inflammatory diseases → may include
 - ✓ Joint, skin, **Heart**
- **Lesion formed**
 - ⇒ Cytokine release, cell recruitment
 - ⇒ The lesion can be in any part of heart & pericardium. As a result, it is called pancarditis



Cont...

B. Endocarditis (usually acute)

- The disease mechanism

→ just like *S. aureus* & others
- ⇒ Bacterial proliferation
 - ✓ Attach & establish growth
- ⇒ Deposition → fibrin & platelet
 - ✓ Interfere phagocytosis & complement
- ⇒ Abscess formation & tissue destruction
 - ✓ Valve destruction → **may cause heart failure**
 - ✓ Can spread to myocardium

Cont...

❖ Diagnosis

- Culture
- Serological test
 - ✓ ASO Ab test (Anti-streptolysin O antibody)
 - ✓ Anti-DNase B antibody

Cont...

1.4. Viridans streptococci

- Gram positive
- Diverse group of bacteria
- No group specific Agn
- Part of normal flora
 - ⇒ Oropharynx, GIT (gastrointestinal tract)
 - ⇒ GUT (genitourinary tract)

Cont...

❖ Pathogenesis

▪ Transmission

⇒ Mostly endogenous source

▪ Virulent factor→

⇒ Have surface proteins

✓ Used for adhesion on host cell

⇒ But lack potent toxin,

▪ Manipulations

⇒ Tooth extraction → Bacteremia → Heart

⇒ Tooth surgery

▪ Others

Cont...

❖ Diseases

- Endocarditis (**usually sub acute**)
- The gradual onset of symptoms
- Can be caused by
 - ⇒ *Streptococcus mutans, Streptococcus mitis, ...*
- Common → if heart valve damaged
 - E.g. Rheumatic fever
 - Prosthetic valve

Cont...

❖ Laboratory diagnosis

⇒ Culture

Cont...

1.5. Enterococcus

- Gram positive → group D carbohydrate
- Part of normal flora
 - ⇒ GIT
- Most non-hemolytic
- Include
 - ⇒ *Enterococcus faecalis*
 - ⇒ *Enterococcus faecium*

Cont...

❖ Pathogenesis

- **Transmission**

- ⇒ Mostly endogenous source

- **Virulent factor**

- ⇒ Have surface proteins

- ✓ Adhesion

- ⇒ Resistant to many antibiotics

- **Do not have**

- ⇒ Potent toxin

- ⇒ Limited potential to cause diseases

Cont...

- But cause diseases
 - ⇒ Immuno compromised individuals
 - ⇒ Hospitalized patients
 - ✓ Nosocomial infection → one of the Common cause
 - ✓ ↑ risk → abdominal surgery, catheter,
 - ⇒ Broad antibiotic treatment
 - ✓ Allow Enterococcus to survive, over grow & colonize

Cont...

- Enter to b'd circulation → bacteremia
 - ⇒ Can spread to heart
 - ⇒ Can lodge on heart valves
 - ✓ Start proliferation
 - ✓ Deposition → fibrin & platelets
 - ✓ Prevent phagocytosis & complement
 - ⇒ Vegetation formed

Cont...

❖ Diseases

- Endocarditis (usually sub acute)
 - ⇒ Common → if valvular disease exists
 - ✓ Become more susceptible

❖ Laboratory diagnosis

- Culture
 - Staphylococcus, Streptococci & Enterococcus
 - ⇒ Cause 85% of infective endocarditis¹

1. Guideline on prevention, diagnosis & treatment of infective endocarditis. **European Heart Journal (2009)**, 30: 2369-2413

Cont...

1.6. Coxiella burnetii

- Coccobacillus
- Gram negative bacteria
 - ⇒ Poorly stain with gram stain
 - ⇒ Can be stained by Giemsa stain
- Obligate intracellular
- Spore former
 - ⇒ Resist harsh environment
 - ⇒ Survive for > a year in soil

Cont...

- Worldwide distribution
- Mainly infect animals
 - ⇒ Zoonotic disease
 - ⇒ Humans **working with animals**
 - ✓ ↑ risk



Cont...

❖ Pathogenesis

■ Transmission

⇒ Inhalation of airborne particles

▶ Feces, urine & Milk drinking

⇒ Person-to-person → b'd transfusion,

■ Virulence factor

⇒ Antigenic phase transition

✓ Phase I to Phase II → On their LPS

⇒ Phagolysosome multiplication → resist lysozyme enzyme

Cont...

- Pathogenesis→ Less understood
 - ⇒ Self-limited infection
 - ✓ Rare fetal case
- **Enter→ phagocytosis→ phagolysosome formation**
 - ⇒ Acidic environment→ metabolic machinery
 - ⇒ Intracellular multiplication

Cont...

- Proliferate in RT (respiratory tract)
 - ⇒ Then dissemination
 - ✓ liver, BM, spleen, **heart**
- Mostly chronic patients

▪ **High Ab titer** detected

⇒ For phase I & II Agn

▪ **Immune complexes** formation

⇒ Responsible for some signs & symptoms

Cont...

❖ Disease

- Q-fever → could be acute or chronic
- Acute infection manifested
 - ⇒ Self limiting → If severe
 - ⇒ Influenza like illness or pneumonia
- Chronic infection manifested as endocarditis
 - ⇒ Mainly on prosthetic/damaged valve
 - ⇒ Long incubation → months to years

Cont...

❖ Laboratory diagnosis

- Serology
- Cell culture
- Molecular techniques
 - ⇒ PCR

Cont...

1.7. *HACEK* group

- A group of bacteria
 - ⇒ Fastidious & slow growing
 - ⇒ Gram negative bacteria
- The group include
 - ⇒ *Haemophilus* spp
 - ⇒ *Actinobacillus actinomycetemcomitans*
 - ⇒ *Cardiobacterium hominis*
 - ⇒ *Eikenella corrodens*
 - ⇒ *Kingella kingae*
- Part of normal flora → **oral cavity**

Cont...

❖ Pathogenesis

▪ *HACEK* endocarditis

▪ Oropharynx

- ⇒ History of oral disease or
- ⇒ Dental procedures ,

Bacteremia→

Adhere & multiply→

Cardiac disease

Cont...

❖ Disease

- Associated with localized infection in mouth
- Can also cause disseminated infection
 - ⇒ Commonly endocarditis
 - ⇒ Can cause in native/prosthetic valve
- ~3% of endocarditis
 - ⇒ By HACEK
 - ⇒ Usually sub acute endocarditis

Cont...

❖ **Laboratory diagnosis**

- **Culture**

 - ⇒ Slow growing → upto 30 days

- **Molecular techniques**

 - ⇒ PCR

- **Further reading**

 - ⇒ Harrison's principle of internal medicine

Cont...

1.8. *Borrelia*

- Thin walled, gram-negative spirochetes
- Weakly stain by gram stain
 - ⇒ Can be stained → Giemsa or Wright stain
- Can cause diseases
 - ⇒ Relapsing fever & Lyme disease
- Include d/t specieses
 - ⇒ *B. burgdorferi*
 - ⇒ Other *Borrelia* species

Cont...

❖ Pathogenesis

- *B. burgdorferi*
- Main reservoir → rodent
 - ⇒ Zoonotic disease
 - ⇒ Transmitted by tick bite
- **Virulent factor**
 - ⇒ Outer surface protein
 - ✓ Adhesion structures
 - ✓ Antigenic shift
- The exact disease mechanism
 - ⇒ Not known

Cont...

❖ Disease

- Lyme disease
 - ⇒ Found in USA, Asia, Europe
- Begins as localized infection
 - E.g. hand skin
- If untreated → disseminated using b'd
 - ⇒ Cause cardiac dysfunction
 - E.g. Myocarditis, pericarditis



Cont...

❖ Diagnosis

- Serology
- Molecular technique → PCR
- Culture
 - ⇒ Slow growing → many days to weeks
 - ⇒ Complex nutritional requirement
 - ✓ Culture not effective

Prevention

- General points
 - ⇒ Aseptic surgery
 - ⇒ Prophylactic antibiotics
 - ✓ Before invasive procedures
 - ✓ Minimize/prevent bacteremia → endocarditis
 - ⇒ Avoid vectors → E.g. Lyme disease
 - ⇒ Minimizing antibiotic resistance
- Many more.....
 - Read more

2. Viruses

2.1. Coxsackie virus

- Family → Picornaviridae (Pico (small) RNA virus)
- Genus → *Enterovirus*
 - ⇒ Human *Enterovirus* → include ≥ 71 serotypes
 - ✓ Polioviruses, **Coxsackie viruses** (A & B) or echoviruses

Cont...

- Cocksackie viruses
 - ⇒ Named by town → Cocksackie (in New York state)
 - ⇒ Two groups
 - ✓ A (1-22, 24) & B (1-16 types)
 - ✓ Based on biologic & antigenic d/nces

Cont...

❖ Pathogenesis of enteroviruses

■ Transmission

- ✓ Direct or indirect fecal-oral
 - ▶ As the name indicates

■ Virulence factor

- ⇒ Resistant to harsh environment → b/c capsid
 - ✓ Acidic PH & some disinfectants (70% alcohol)

■ Have difference in

- ✓ Tissue tropism & cytolytic capacity

Cont...

- **Enteroviruses** replication
 - ⇒ Replication initiation → mucosa & lymphoid tissue
 - ✓ Pharynx & tonsils
 - ⇒ Spread to Peyer's patches
 - ✓ Underlying intestinal mucosa
 - ⇒ Viremia spreads virus
 - ✓ Target tissue with receptors
 - ✓ Secondary replication → secondary viremia
 - ⇒ Most cytolytic
 - ✓ Directly damage target cell

Local replication →

Viremia →

Target cell →

Replication

Cont...

❖ **Disease caused by Coxsackie B viruses**

- **Myocarditis & pericarditis**

- ⇒ Can occur → older children & adults

- ⇒ Most threatening → in **newborns**

- ✓ **Febrile illnesses**

- ▶ Sudden & unexplained heart failure

- ⇒ High mortality

Cont...

❖ **Laboratory diagnosis** Coxsackie B viruses

- Cell culture
- Serology
 - ⇒ Used but challenging
 - ⇒ Because of many serotypes
- Molecular techniques
 - ⇒ RT-PCR
- Prevention
 - ⇒ Personal hygiene
 - ⇒ Proper feces disposal,

Cont...

2.2. Orthomyxoviruses

- The family Orthomyxoviridae include
 - ⇒ Enveloped **Influenza A & B** viruses
 - ⇒ Segmented negative-sense RNA
- Aerosol spread
- Respiratory tract infection
 - ⇒ Rarely spread outside lung
- The **complication** of influenza virus
 - ⇒ Cardiac involvement

⇒ [Reading more...](#)

3. Parasites

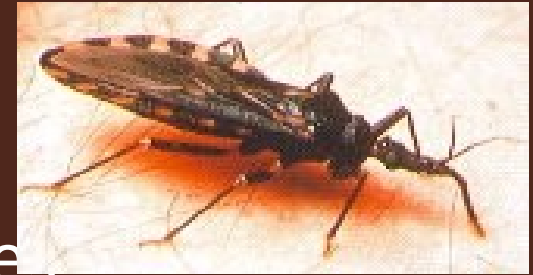
3.1. *Trypanosoma cruzi*

- Flagellate parasite
 - ⇒ Trichomonas, Giardia, Leishmania, Trypanosoma
- Hemoflagellate → Leishmania, Trypanosoma
 - ⇒ Require intermediate insect host
- *T. cruzi* found
 - ⇒ In b'd & tissues (**heart**, muscle,)

Cont...

❖ Pathogenesis

- Transmission → *triatomine* insects
 - ⇒ Release metacyclic trypomastigotes
 - ✓ With feces
 - ⇒ Skin break/conjunctiva/mucosa → enter
 - ✓ Changed into amastigote → (round & no flagella)
- Multiply locally inside cell
 - ⇒ Released to b'd → as trypomastigotes



Trypomastigotes in b'd??

Epimastigotes in triatomine (mul)
→ metacyclic trypomastigotes

Or Infect new cell → amastigote

Cont...

- The cellular/tissue damage → is due to
 - ⇒ Pseudocyst ruptures
 - ✓ Inflammatory reaction occurs
 - ⇒ Parasitic enzyme
 - ⇒ Parasitic Agn → released
 - ✓ Deposited on host tissue
 - ✓ Enhance host tissue distraction
 - ▶ Self tissue directed immune response
- Eg. Endocardium tissue & others

Cont...

❖ Disease

- American trypanosomiasis (Chagas ds)
 - ⇒ Acute & chronic form
- Began → as localized infection
 - ⇒ Skin chagoma formed
 - including d/t cells
 - ⇒ If conjunctiva entry
 - **Romana's sign** (edema)
- Few of the patients
 - ⇒ Develop chronic infection
 - ✓ Involve d/t cells → **mainly heart**

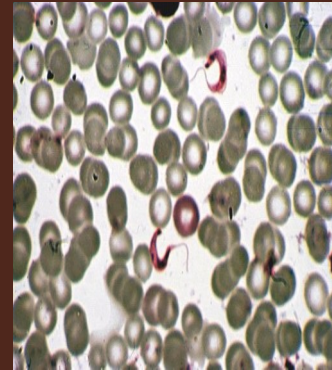


Cont...

- Cause **myocarditis**
 - ⇒ With/without heart enlargement

- ❖ **Laboratory diagnosis**

- Direct microscopy
 - Unstained/stained BF → trypomastigotes
- Serology (chronic)
- Xenodiagnosis



Starve *triatomine* →

Allow bite →

~25-30 days examine feces

- Molecular techniques → PCR

4. Fungus

- Fungus
 - ⇒ Rarely cause heart disease
 - ⇒ But some fungi can be involved
 - ✓ Candida spp
 - ✓ Aspergillus spp
 - ⇒ Usually occur
 - ✓ Prosthetic valve
 - ✓ Cardiac valve surgery
 - ✓ Immune compromised

■ Reading assignment

Infective Endocarditis ...

ORGANISMS CAUSING MAJOR CLINICAL FORMS OF ENDOCARDITIS

ORGANISM	PERCENT OF CASES							
	NATIVE VALVE ENDOCARDITIS		PROSTHETIC VALVE ENDOCARDITIS AT INDICATED TIME OF ONSET (MONTHS) AFTER VALVE SURGERY			ENDOCARDITIS IN INJECTION DRUG USERS		
	COMMUNITY-ACQUIRED (n = 683)	HEALTH CARE-ASSOCIATED (n = 128)	<2 (n = 144)	2-12 (n = 31)	>12 (n = 194)	RIGHT-SIDED (n = 346)	LEFT-SIDED (n = 204)	TOTAL (n = 675) ^a
Streptococci ^b	32	8	1	9	31	5	15	12
Pneumococci	1	—	—	—	—	—	—	—
Enterococci	8	16	8	12	11	2	24	9
<i>Staphylococcus aureus</i>	35	44 ^c	22	12	18	77	23	57
Coagulase-negative staphylococci	4	15	33	32	11	—	—	—
Fastidious gram-negative coccobacilli (HACEK group) ^d	3	—	—	—	6	—	—	—
Gram-negative bacilli	3	5	13	3	6	5	13	7
<i>Candida</i> spp.	1	6	8	12	1	—	12	4
Polymicrobial/miscellaneous	6	1	3	6	5	8	10	7
Diphtheroids	—	—	6	—	3	—	—	0.1
Culture-negative	5	5	5	6	8	3	3	3

Lecture summery

Microbes affecting heart & pericardium

1. Bacteria

- ⇒ *S. aureus*
- ⇒ *S. epidermidis*
- ⇒ *S. pyogenes*
- ⇒ Viridians streptococcus
- ⇒ Enterococcus
- ⇒ *Coxiella burnetii*
- ⇒ HACEK Group
- ⇒ *B. burgdorferi*

2. Virus

- ⇒ Enterovirus
 - ✓ Coxsackie Virus

3. Parasite

- ⇒ *T. cruzi*

4. Fungus

- ⇒ *Candida* spp
- ⇒ *Aspergillus* spp